

Task 1

This task aims to perform Market Basket Analysis using the Apriori algorithm on a retail transaction dataset.

The dataset contains customer transactions. Each row represents a transaction, and each transaction includes multiple purchased items.

The task includes the following steps:

1. Load and preprocess the dataset.
2. Apply the Apriori algorithm to extract frequent itemsets.
3. Generate association rules using support, confidence, and lift.
4. Sort rules based on lift to identify strong associations.
5. Export frequent itemsets and association rules into CSV files.
6. Visualize the strongest associations using appropriate charts.

Note:

The goal is to identify frequently purchased product combinations and discover meaningful relationships between items.